EE3001: Solid State Devices
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Assignment #4

Try yourself to solve the problems with detailed steps. Take a limited time quiz to test your understanding on related concepts.

Problems will be discussed in class. You need to contact TAs or teacher if you are unable to do anything useful by yourself or with the help of your friends.

If needed, assume the following values by default unless otherwise specified: Si is the default semiconductor; SiO_2 is the default oxide; T @room temperature is 300 K; n_i**@room temp. =** $10^{10} cm^{-3}$**;** k_BT**=0.026 eV (at room temperature);** $E_{G,Si}$ **= 1.12 eV and temperature independent; at 300 K,** N_c**=**$2.8 \times 10^{19} cm^{-3}$ **and** N_v**=**$1.02 \times 10^{19} cm^{-3}$

1. (5 points) Light is shone on the surface of an n-type silicon wafer generating 10^{16} electron-hole pairs per cm^2 per second only on this surface. Assuming T = 300K, μ_n= 1350 cm^2/V-sec, μ_p= 450 cm^2/V-sec, and τ_p=τ_n=1 μs what is the electron current density (in mA/cm^2), at a distance of 5 μm from this surface?

2. (5 points) Fig.1 shows a non-uniformly doped n-type semiconductor block with 3 distinct regions. For $x < 0$, $N_D = 10^{14}$ $/cm^3$ and for $x > l$, $N_D = 10^{18}$ $/cm^3$. For $0 < x < l$, the donor concentration increases exponentially and can be expressed as

   $$N_D(x) = N_{D0} e^{ax},$$

   where N_{D0} and a are constants. Assuming $l = 1$ μm, plot the electric field variation in the semiconductor block at room temperature (300 K).

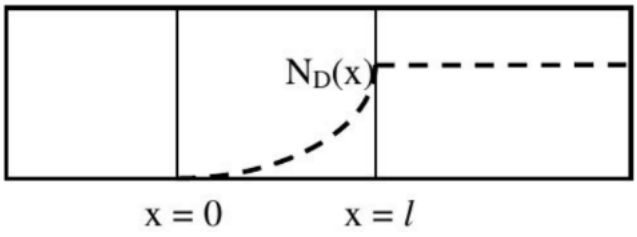


Figure 1: A bar of n-type semiconductor showing graded impurity distribution.

3. (5 points) Fig. 2 shows a sample used for the Hall Effect experiment, where $W = 0.1$ mm, $d = 1$ mm and $L = 1$ cm. A current of $I = 1$ A flows through the sample when $V_{CD} = 50$ V. If the measured Hall voltage is $V_H = -6.25$ mV when the current flowing through the sample is $I = 1$ A and the applied magnetic field strength is $B_z = 10^{-5}$ Wb/cm^2 , what is

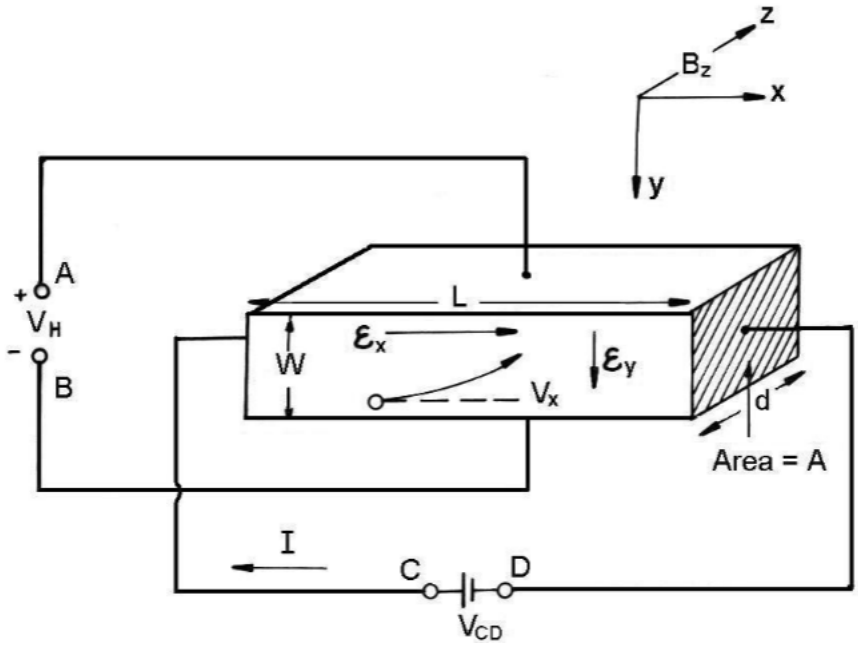


Figure 2: Hall experiment setup.

(a) the type of the semiconductor,

(b) the doping concentration (assuming all dopants are ionized) and

(c) the mobility of the majority carriers in the sample?